

DSA0502 – Query Processing for Data Science	
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LAB PROGRAMS 29 AND 30

29. Display a Bar Chart with Different Colors for Each Bar

Question

Write a Python program to display a bar chart of the popularity of programming languages. Use different colors for each bar.

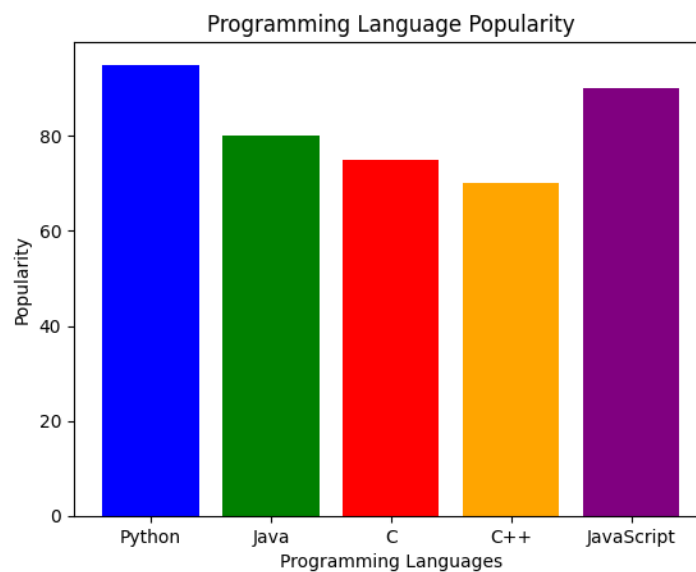
Aim

To display a colorful bar chart by assigning different colors to each programming language.

Code

```
import matplotlib.pyplot as plt
languages = ["Python", "Java", "C", "C++", "JavaScript"]
popularity = [95, 80, 75, 70, 90]
colors = ["blue", "green", "red", "orange", "purple"]
plt.bar(languages, popularity, color=colors)
plt.xlabel("Programming Languages")
plt.ylabel("Popularity")
plt.title("Programming Language Popularity")
plt.show()
```

Output



Result

A colorful bar chart was created successfully using different colors for each bar.

30. Create a Bar Plot of Scores by Group and Gender

Question

Write a Python program to create a bar plot of scores by group and gender. Use multiple X values on the same chart for men and women.

Aim

To compare the scores of men and women using a grouped bar chart.

Dataset

```
groups = ["Group A", "Group B", "Group C", "Group D"]
```

```
men = [22, 30, 35, 35]
```

```
women = [25, 32, 30, 35]
```

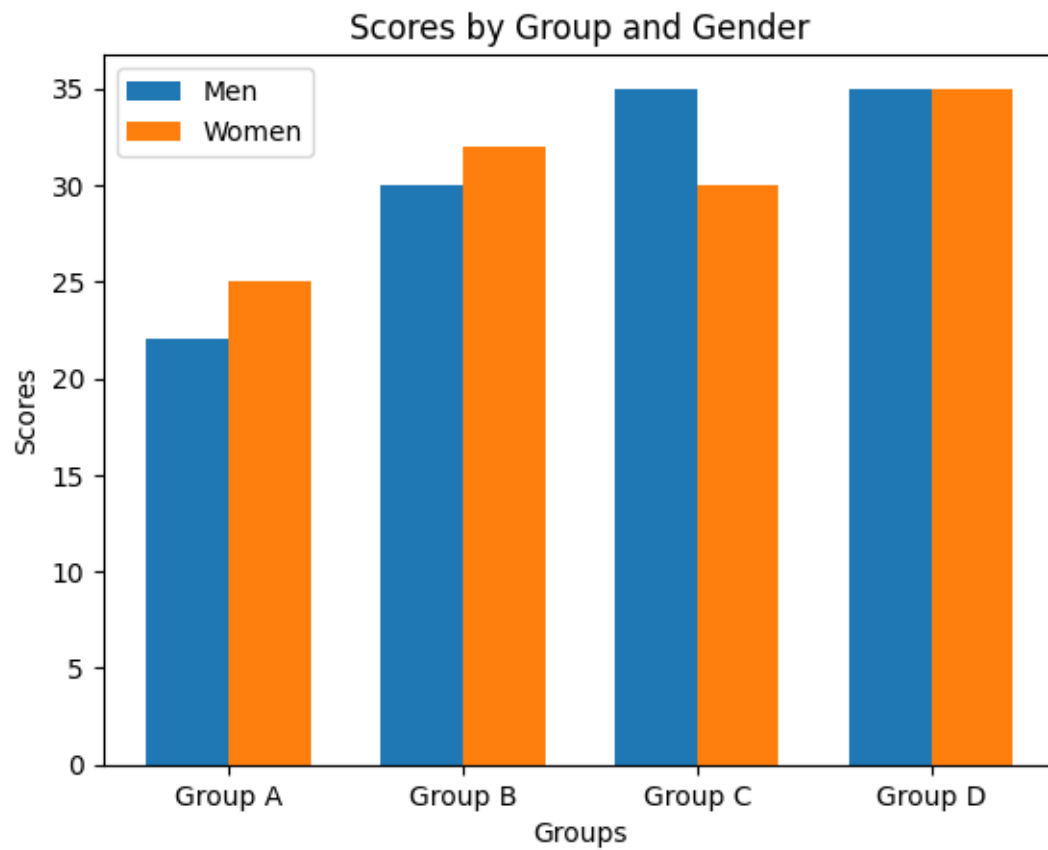
Code

```
import matplotlib.pyplot as plt
import numpy as np

groups = ["Group A", "Group B", "Group C", "Group D"]
men = [22, 30, 35, 35]
women = [25, 32, 30, 35]
x = np.arange(len(groups))
width = 0.35

plt.bar(x - width/2, men, width, label="Men")
plt.bar(x + width/2, women, width, label="Women")
plt.xticks(x, groups)
plt.xlabel("Groups")
plt.ylabel("Scores")
plt.title("Scores by Group and Gender")
plt.legend()
plt.show()
```

Output



Result

The scores of men and women were compared successfully using a grouped bar chart.